

Calibration Under Shift Reading Synthesis

Research reading artifact | fictional/internal | 2026-07-13

Analytical frame. HF-Eval-S1/S2 v3.2 is a fictional 2,400-record benchmark with 1,600 S1 and 800 S2 records. The calibration lens is applied to the 2026-07-13 Helix Forge Lab R184 snapshot, where Model Alpha M-14 has F1 0.873, without asserting any real population shift.

Reliability-bin reading

Confidence bins compare stated belief with synthetic correctness. A bin is useful when its count and observed rate travel together in the review record.

ECE interpretation

Expected calibration error summarizes weighted bin gaps. A lower value is not automatically safer if a critical label has sparse support.

Partition-specific risks

S1 can look stable while S2 exposes confidence compression in route-versus-resolve examples. Threshold changes require a partition-specific rationale.

Working matrix

Partition / bin	Records	Observed accuracy	Reading
S1: 0.8-0.9	420	0.86	Slight overconfidence
S2: 0.8-0.9	140	0.76	Review shift
S2: 0.5-0.6	95	0.58	Near reliable

S1 / S2 reliability-bin comparison

Bin	S1 count	S1 accuracy	S2 accuracy	Interpretation
0.9-1.0	360	0.92	0.82	High-confidence shift
0.8-0.9	420	0.86	0.76	Overconfidence grows
0.5-0.6	190	0.57	0.58	Near reliable

Mitigation actions

1. Refit no threshold until S2 bin counts are checked. 2. Sample route/resolve cases in the shifted high-confidence band. 3. Record ECE beside the bin table, not in isolation.

Traceable readings

The comparison is derived from *Partition Confidence Ledger*, *ECE Interpretation Card*, and *S2 Stress-Slice Memo* [synthetic/internal].